

Lecture 5: Hypothesis testing

Economics 527: Econometric Methods

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Hypotheses and errors

Estimator and parameter set

- Normally distributed estimator with known variance ω^2 :

$$\hat{\theta}_{1 \times 1} \sim N\left(\theta_{1 \times 1}, \omega_{1 \times 1}^2\right), \quad \frac{\hat{\theta} - \theta}{\omega} = Z \sim N(0, 1)$$

- Regression example, conditional on X :

$$\theta = \beta_1, \hat{\theta} = \hat{\beta}_1, \omega^2 = \sigma^2 / (X_1^\top M_2 X_1).$$

- Parameter set, for example $\Theta = \mathbb{R}$:

$$\theta \in \Theta$$

- Two disjoint parts:

$$\begin{aligned} \Theta &= \Theta_0 \cup \Theta_1 \\ \Theta_0 \cap \Theta_1 &= \emptyset \end{aligned}$$

- Null and alternative hypotheses:

$$H_0 : \theta \in \Theta_0 \quad \text{vs} \quad H_1 : \theta \in \Theta_1$$

Examples of Θ_0 and Θ_1

- Point null, $\Theta = \mathbb{R}$:

$$\begin{aligned} \Theta_0 &= \{0\} \\ \Theta_1 &= \mathbb{R} \setminus \{0\} \\ \Leftrightarrow \quad H_0 : \theta &= 0 \quad \text{vs} \quad H_1 : \theta \neq 0 \end{aligned}$$

- More generally, for a **known** θ_0 :

$$\begin{aligned} \Theta_0 &= \{\theta_0\} \\ \Theta_1 &= \mathbb{R} \setminus \{\theta_0\} \\ \Leftrightarrow \quad H_0 : \theta &= \theta_0 \quad \text{vs} \quad H_1 : \theta \neq \theta_0 \end{aligned}$$

- One-sided null:

$$\begin{aligned} \Theta_0 &= [\theta_0, +\infty) \\ \Theta_1 &= (-\infty, \theta_0) \\ \Leftrightarrow \quad H_0 : \theta &\geq \theta_0 \quad \text{vs} \quad H_1 : \theta < \theta_0 \end{aligned}$$

- **Definition.** Simple hypothesis: a single-point set; otherwise composite:

$$\Theta_0 = \{\theta_0\} : \text{simple}, \quad \Theta_0 = [0, +\infty) : \text{composite}$$

Test statistic and decision rule

- Test statistic T : a function of the data, random, with range S :

$$\begin{aligned} T &\in S \\ S &= S_A \cup S_R, \quad S_A \cap S_R = \emptyset \end{aligned}$$

- S_A : acceptance region; S_R : rejection region.
- Decision rule:

$$\begin{aligned} T \in S_A &\implies \text{decide in favor of } \theta \in \Theta_0 && (\text{fail to reject } H_0) \\ T \in S_R &\implies \text{decide in favor of } \theta \in \Theta_1 && (\text{reject } H_0) \end{aligned}$$

Type I and Type II errors

Truth against decision:

Truth	Decision H_0	Decision H_1
H_0	correct	Type I error
H_1	Type II error	correct

- Probabilities of the errors:

$$\begin{aligned} \text{P(Type I error)} &= \text{P}(T \in S_R \mid \theta \in \Theta_0) \\ \text{P(Type II error)} &= \text{P}(T \notin S_R \mid \theta \in \Theta_1) \\ &= \text{P}(T \in S_A \mid \theta \in \Theta_1) \end{aligned}$$

- To make $\text{P}(\text{Type I error})$ smaller:

$$\begin{aligned} &S_R \downarrow \\ \implies &S_A \uparrow \\ \implies &\text{P}(\text{Type II error}) \uparrow \end{aligned}$$

- We can't make both error probabilities small at the same time.

Valid test, size, significance level

- **Definition.** Valid test, with α chosen in advance:

$$\begin{aligned} \text{P}(\text{Type I error}) &= \text{P}(\text{falsely deciding in favour of } H_1 \mid H_0 \text{ true}) \\ &\leq \alpha \end{aligned}$$

- Composite Θ_0 , supremum of the Type I error probabilities:

$$\sup_{\theta \in \Theta_0} \text{P}(T \in S_R \mid \theta) \leq \alpha$$

- Size α test:

$$\sup_{\theta \in \Theta_0} \text{P}(T \in S_R \mid \theta) = \alpha$$

- Significance level α : small, for example 0.01, 0.05, 0.1.

Two-sided test

Point null against a two-sided alternative

- Hypotheses, for example $\theta_0 = 0$:

$$\underbrace{H_0 : \theta = \theta_0}_{\text{simple}} \quad \text{vs} \quad \underbrace{H_1 : \theta \neq \theta_0}_{\text{composite}}$$

- Three objects:

θ : true value, unknown, fixed

$\hat{\theta}$: estimator, data, random

θ_0 : chosen by you, known, fixed

- Regression example, $\theta = \beta_1$ and $\theta_0 = 0$:

$$H_0 : \beta_1 = 0 \quad \text{vs} \quad H_1 : \beta_1 \neq 0$$

Test statistic $T(\theta_0)$

- Distance between the data and H_0 :

$$T(\theta_0) = \left| \frac{\hat{\theta} - \theta_0}{\omega} \right|$$

- Reject H_0 when:

$$\begin{aligned} & T(\theta_0) \text{ is large} \\ \iff & T(\theta_0) > c \end{aligned}$$

- Regions, $c > 0$:

$$S_R = (c, +\infty)$$

$$S_A = [0, c]$$

$$S = S_A \cup S_R = [0, +\infty)$$

- Choose c for size control:

$$P(\text{Type I error}) \leq \alpha$$

Choice of c

- Size α : set the Type I error probability at $\theta = \theta_0$ to α :

$$\begin{aligned}
 \alpha &= P(\text{Type I error}) \\
 &= P(\text{reject } H_0 \mid H_0 \text{ true}) \\
 &= P(T(\theta_0) > c \mid \theta = \theta_0) \\
 &= P\left(\left|\frac{\hat{\theta} - \theta_0}{\omega}\right| > c \mid \theta = \theta_0\right) \\
 &\stackrel{\theta = \theta_0}{=} P\left(\underbrace{\left|\frac{\hat{\theta} - \theta}{\omega}\right|}_{= Z \sim N(0,1)} > c\right) \\
 &= P(|Z| > c) \\
 &= P(\underbrace{\{Z > c\}}_A \text{ or } \underbrace{\{-Z > c\}}_B) \\
 &= P(Z > c) + P(Z < -c) \\
 &= 2P(Z > c) \\
 \Rightarrow P(Z > c) &= \frac{\alpha}{2} \\
 \Rightarrow P(Z \leq c) &= 1 - \frac{\alpha}{2} \\
 \Rightarrow c &= z_{1-\alpha/2} \quad (P(Z \leq z_\tau) = \tau)
 \end{aligned}$$

Size α two-sided test

- Size α test of $H_0 : \theta = \theta_0$ against $H_1 : \theta \neq \theta_0$:

Reject H_0 if $\left|\frac{\hat{\theta} - \theta_0}{\omega}\right| > z_{1-\alpha/2}$

- For $\alpha = 0.05$:

$$\begin{aligned}
 z_{0.975} &\approx 1.96 \approx 2 \\
 \Rightarrow \text{reject if } \left|\frac{\hat{\theta} - \theta_0}{\omega}\right| &\gtrsim 2 \\
 \Leftrightarrow \text{reject if } |\hat{\theta} - \theta_0| &\gtrsim 2 \underbrace{\omega}_{\text{std. dev.}}
 \end{aligned}$$

- Compare with: Reject if $|\hat{\theta} - \theta_0| > 5\omega$:

$$P(Z > 5) \approx 3 \cdot 10^{-7}, \quad \alpha = P(|Z| > 5) \approx 6 \cdot 10^{-7}$$

Significance at level α

- Definition.** Significance at level α , by a size α test:

$$\begin{aligned}
 \left|\frac{\hat{\theta}}{\omega}\right| > z_{1-\alpha/2} &\Rightarrow H_0 : \theta = 0 \text{ rejected in favour of } H_1 : \theta \neq 0 \\
 &\Rightarrow \hat{\theta} \text{ significant at level } \alpha
 \end{aligned}$$

Test and confidence interval

- Solve for θ_0 :

$$\begin{aligned}
 H_0 \text{ rejected} &\iff \left| \frac{\hat{\theta} - \theta_0}{\omega} \right| > z_{1-\alpha/2} \\
 H_0 \text{ not rejected} &\iff \left| \frac{\hat{\theta} - \theta_0}{\omega} \right| \leq z_{1-\alpha/2} \\
 &\iff -z_{1-\alpha/2} \leq \frac{\hat{\theta} - \theta_0}{\omega} \leq z_{1-\alpha/2} \\
 &\iff -z_{1-\alpha/2} \omega \leq \hat{\theta} - \theta_0 \leq z_{1-\alpha/2} \omega \\
 &\iff z_{1-\alpha/2} \omega \geq \theta_0 - \hat{\theta} \geq -z_{1-\alpha/2} \omega \\
 &\iff \hat{\theta} + z_{1-\alpha/2} \omega \geq \theta_0 \geq \hat{\theta} - z_{1-\alpha/2} \omega \\
 &\iff \hat{\theta} - z_{1-\alpha/2} \omega \leq \theta_0 \leq \hat{\theta} + z_{1-\alpha/2} \omega \\
 &\iff \theta_0 \in CI_{1-\alpha} = [\hat{\theta} - z_{1-\alpha/2} \omega, \hat{\theta} + z_{1-\alpha/2} \omega]
 \end{aligned}$$

- Equivalent statement of the test:

Reject $H_0 : \theta = \theta_0$ if $\theta_0 \notin CI_{1-\alpha}$

$CI_{1-\alpha}$ as a set of null values

- All θ_0 the test does not reject:

$$\begin{aligned}
 CI_{1-\alpha} &= \{\theta_0 : T(\theta_0) \leq z_{1-\alpha/2}\} \\
 &= \left\{ \theta_0 : \left| \frac{\hat{\theta} - \theta_0}{\omega} \right| \leq z_{1-\alpha/2} \right\} \\
 &= \{\theta_0 : \text{a size } \alpha \text{ test does not reject } H_0 : \theta = \theta_0\}
 \end{aligned}$$

- Suppose in your data:

$$CI_{0.95} = [0.04, 0.06]$$

Power

Noise and signal

- Type II error:

$$P(\text{Type II error}) = P(\text{fail to reject } H_0 \mid \theta \neq \theta_0)$$

- Split $(\hat{\theta} - \theta_0)/\omega$, under the true θ :

$$\begin{aligned}
 \underbrace{\frac{\hat{\theta} - \theta_0}{\omega}}_{\text{data} - H_0} &= \underbrace{\frac{\hat{\theta} - \theta}{\omega}}_{= Z \sim N(0,1), \text{ noise}} + \underbrace{\frac{\theta - \theta_0}{\omega}}_{\text{truth} - H_0, \text{ signal}} \\
 &= Z + \frac{\theta - \theta_0}{\omega} \\
 &\sim N\left(\frac{\theta - \theta_0}{\omega}, 1\right)
 \end{aligned}$$

Power function $\pi(\theta)$

- Probability of rejecting H_0 , as a function of the true θ :

$$\begin{aligned}
 \pi(\theta) &= P(\text{reject } H_0 \mid \theta) \\
 &= P\left(\left|\frac{\hat{\theta} - \theta_0}{\omega}\right| > z_{1-\alpha/2} \mid \theta\right) \\
 &= P\left(\left|\frac{\hat{\theta} - \theta}{\omega} + \frac{\theta - \theta_0}{\omega}\right| > z_{1-\alpha/2} \mid \theta\right) \\
 &= P\left(\left|\tilde{Z} + \frac{\theta - \theta_0}{\omega}\right| > z_{1-\alpha/2}\right)
 \end{aligned}$$

- Known: θ_0 , ω , $z_{1-\alpha/2}$, the distribution of Z .
- Unknown: $\theta \in \mathbb{R}$.

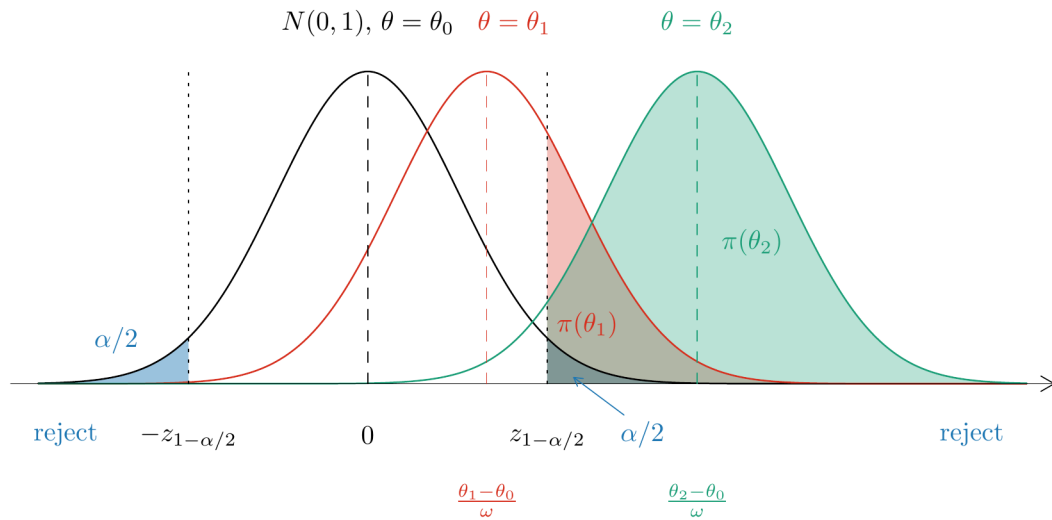
Type II error probability

- Type II error probability depends on θ : For $\theta \neq \theta_0$,

$$\begin{aligned}
 P(\text{Type II error} \mid \theta) &= P(\text{fail to reject } H_0 \mid \theta) \\
 &= 1 - P(\text{reject } H_0 \mid \theta) \\
 &= 1 - \pi(\theta).
 \end{aligned}$$

Distribution of $(\hat{\theta} - \theta_0)/\omega$ under H_0 and H_1

- Density of $(\hat{\theta} - \theta_0)/\omega$, with $\theta_0 < \theta_1 < \theta_2$:

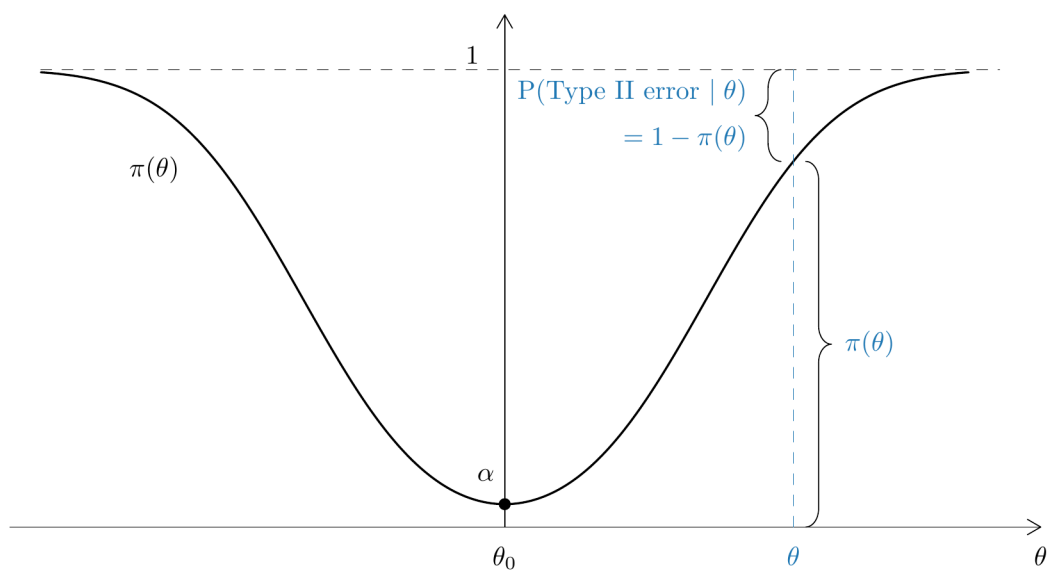


- Under $\theta \neq \theta_0$, more mass in the rejection region:

$$P\left(\left|\frac{\hat{\theta} - \theta_0}{\omega}\right| > z_{1-\alpha/2} \mid \theta\right) > \alpha$$

Power curve

- $\pi(\theta)$ against θ : minimum α at θ_0 , approaches 1 far from θ_0 :



- Largest Type II error probability, as $\theta \rightarrow \theta_0$:

$$\begin{aligned} \sup_{\theta \neq \theta_0} P(\text{Type II error} \mid \theta) &= \sup_{\theta \neq \theta_0} P\left(\left|\frac{\hat{\theta} - \theta_0}{\omega}\right| \leq z_{1-\alpha/2} \mid \theta\right) \\ &= \sup_{\theta \neq \theta_0} P\left(\left|Z + \frac{\theta - \theta_0}{\omega}\right| \leq z_{1-\alpha/2}\right) \\ &= 1 - \alpha \end{aligned}$$

Maximum against supremum

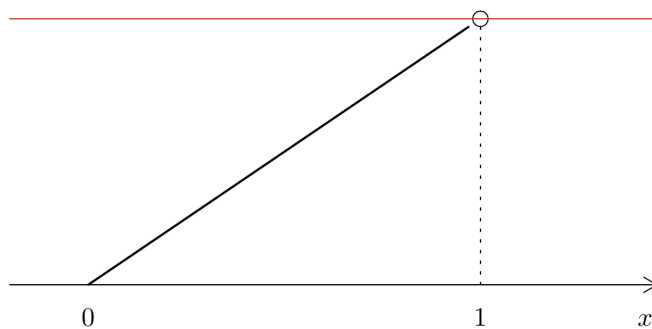
- Closed interval:

$$\max_{x \in [0,1]} x = 1$$

- Half-open interval:

$$\max_{x \in [0,1)} x \text{ does not exist}$$

$$\sup_{x \in [0,1)} x = 1$$



- **Definition.** Supremum:

$$\sup_{x \in \mathcal{X}} f(x) = \text{smallest upper bound of } \{f(x) : x \in \mathcal{X}\}$$

- $P(\text{Type II error} \mid \theta)$ over $\theta \neq \theta_0$: no maximum.

Example: $\hat{\theta} \sim N(\theta, 9)$

- $H_0 : \theta = 0$ vs $H_1 : \theta \neq 0$, $\alpha = 0.05$, $\omega = 3$:

$$\text{Reject } H_0 \text{ if } \left| \frac{\hat{\theta}}{3} \right| > 1.96$$

- Power function:

$$\begin{aligned} \pi(\theta) &= P\left(\left| \frac{\hat{\theta}}{3} \right| > 1.96 \mid \theta\right) \\ &= P\left(\left| \frac{\hat{\theta} - \theta}{3} + \frac{\theta}{3} \right| > 1.96\right) \\ &= P\left(\left| Z + \frac{\theta}{3} \right| > 1.96\right) \\ &= P\left(Z + \frac{\theta}{3} > 1.96 \text{ or } Z + \frac{\theta}{3} < -1.96\right) \\ &= P\left(Z > 1.96 - \frac{\theta}{3}\right) + P\left(Z < -1.96 - \frac{\theta}{3}\right) \end{aligned}$$

Example, power at $\theta = 3$

- Substitute $\theta = 3$:

$$\begin{aligned} \pi(3) &= P\left(Z > 1.96 - \frac{3}{3}\right) + P\left(Z < -1.96 - \frac{3}{3}\right) \\ &= P(Z > 0.96) + P(Z < -2.96) \\ &\approx \underbrace{P(Z \leq -0.96)}_{\approx 0.16853} + 0.00154 \\ &\approx 0.17 \end{aligned}$$

- $\pi(\theta)$ at four values of θ :

$$\pi(\theta) \approx \begin{cases} 0.063 & \theta = -1 \\ 0.05 & \theta = 0 \\ 0.063 & \theta = 1 \\ 0.17 & \theta = 3 \end{cases}$$

Failing to reject $\beta_1 = 0$

- Regression, $\theta = \beta_1$, $\omega^2 = \sigma^2 / (X_1^\top M_2 X_1)$:

$$Y = \beta_1 X_1 + X_2 \beta_2 + U, \quad H_0 : \beta_1 = 0 \quad \text{vs} \quad H_1 : \beta_1 \neq 0$$

- Suppose H_0 is not rejected:

$$\left| \frac{\hat{\beta}_1}{\omega} \right| \leq z_{1-\alpha/2} \quad \stackrel{?}{\implies} \quad \beta_1 = 0$$

- Type II error probability near $1 - \alpha$ for θ near θ_0 .
- $T(\theta_0) > z_{1-\alpha/2}$: strong evidence that $\theta \neq \theta_0$.
- **Claim.** $T(\theta_0) \leq z_{1-\alpha/2}$: not strong evidence in favour of θ_0 .

Minimum detectable deviation

- Detectable θ : $\pi(\theta) \geq \beta$, for example $\beta = 0.85$ or 0.5 .
- Smallest detectable $\bar{\theta} > \theta_0$ solves:

$$P\left(\left|Z + \frac{\bar{\theta} - \theta_0}{\omega}\right| > z_{1-\alpha/2}\right) = \beta, \quad \theta_0, \omega, z_{1-\alpha/2}, \beta \text{ known}$$

- $|\bar{\theta} - \theta_0|$: the minimum detectable deviation.

More precise estimator, more power

- Smaller variance:

$$\begin{aligned} \hat{\theta}^* &\sim N(\theta, \omega_*^2) \\ \omega_*^2 &< \omega^2 \\ \implies \omega_* &< \omega \end{aligned}$$

- Both tests valid:

$$\underbrace{\left|\frac{\hat{\theta} - \theta_0}{\omega}\right| > z_{1-\alpha/2}}_{\text{test with } \hat{\theta}} \quad \text{or} \quad \underbrace{\left|\frac{\hat{\theta}^* - \theta_0}{\omega_*}\right| > z_{1-\alpha/2}}_{\text{test with } \hat{\theta}^*}$$

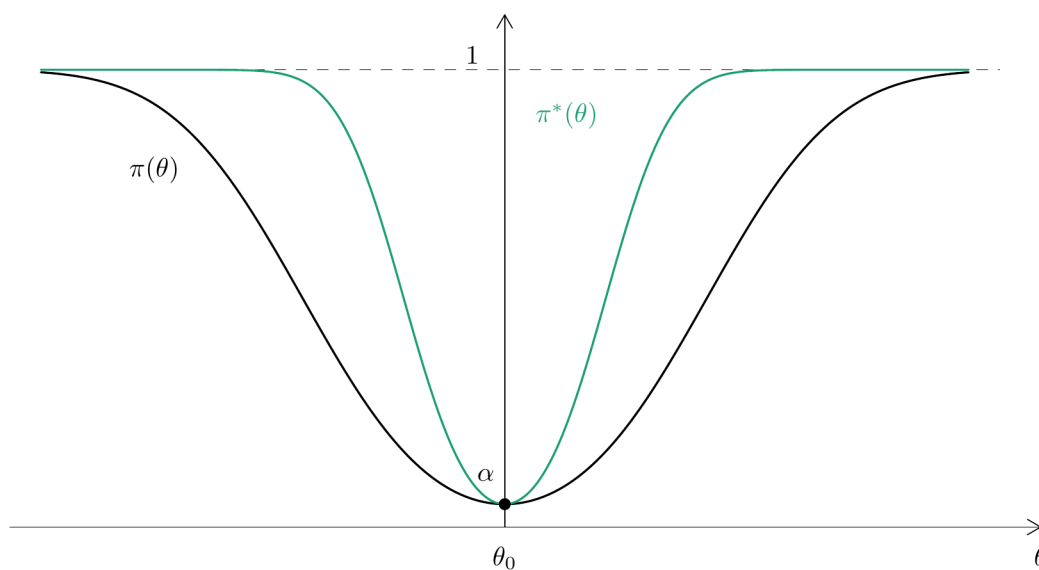
- Suppose $\theta > \theta_0$, larger signal with $\hat{\theta}^*$:

$$\begin{aligned} \frac{\hat{\theta} - \theta_0}{\omega} &= Z + \frac{\theta - \theta_0}{\omega} \sim N\left(\frac{\theta - \theta_0}{\omega}, 1\right) \\ \frac{\hat{\theta}^* - \theta_0}{\omega_*} &\sim N\left(\frac{\theta - \theta_0}{\omega_*}, 1\right) \\ \omega_* < \omega &\implies \frac{\theta - \theta_0}{\omega} < \frac{\theta - \theta_0}{\omega_*} \end{aligned}$$

- Power functions:

$$\begin{aligned} \pi(\theta) &= P\left(\left|\frac{\hat{\theta} - \theta_0}{\omega}\right| > z_{1-\alpha/2} \mid \theta\right) \\ \pi^*(\theta) &= P\left(\left|\frac{\hat{\theta}^* - \theta_0}{\omega_*}\right| > z_{1-\alpha/2} \mid \theta\right) \end{aligned}$$

- Test with $\hat{\theta}^*$: more powerful, $\pi^*(\theta) \geq \pi(\theta)$.



Test that ignores the data

- $V \sim \text{Uniform}(0, 1)$, density and CDF:

$$\begin{aligned}
 f(v) &= \mathbf{1}\{0 < v < 1\} \\
 &= \begin{cases} 1 & 0 < v < 1 \\ 0 & \text{otherwise} \end{cases} \\
 F(v) &= \begin{cases} 0 & v \leq 0 \\ v & 0 < v < 1 \\ 1 & v \geq 1 \end{cases}
 \end{aligned}$$

- For $0 < x < 1$:

$$\begin{aligned}
 P(V < x) &= P(V \leq x) \quad (\text{continuous}) \\
 &= F(x) \\
 &= \int_0^x f(v) dv \\
 &= \int_0^x 1 dv \\
 &= x
 \end{aligned}$$

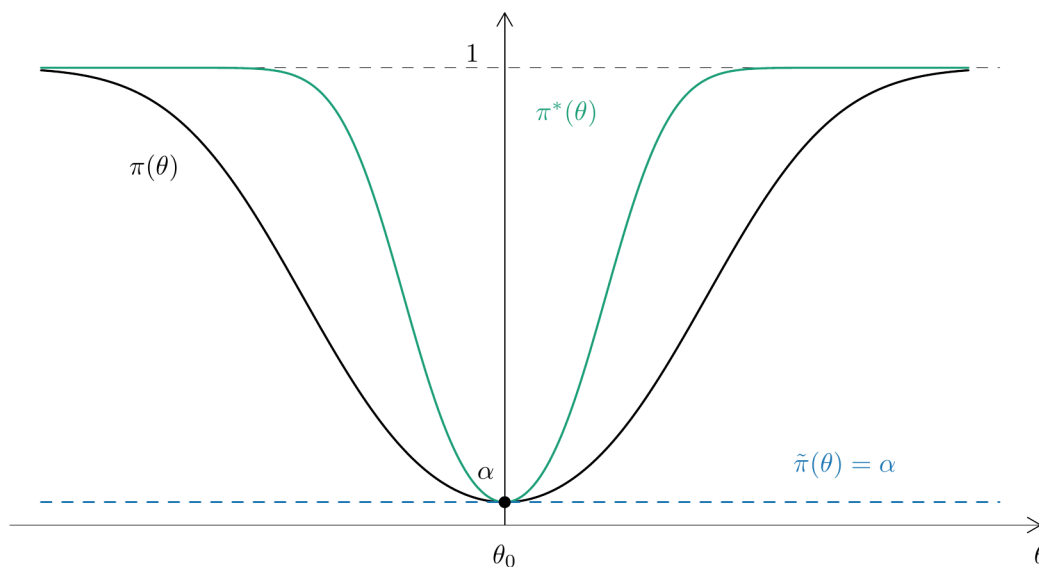
- Test:

1. Throw $\hat{\theta}$ away.
2. Draw $V \sim \text{Uniform}(0, 1)$.
3. Reject $H_0 : \theta = \theta_0$ if $V < \alpha$.

- V does not depend on θ :

$$\begin{aligned}
 P(\text{Type I error} \mid \theta = \theta_0) &= P(V < \alpha) \\
 &= \alpha \\
 \tilde{\pi}(\theta) &= P(\text{reject } H_0 \mid \theta) \\
 &= P(V < \alpha) \\
 &= \alpha \quad \text{for every } \theta
 \end{aligned}$$

- Power α : trivial.



One-sided tests

One-sided hypotheses

- One-sided null and alternative, for example $\theta_0 = 0$:

$$\underbrace{H_0 : \theta \leq \theta_0}_{\text{composite}} \quad \text{vs} \quad \underbrace{H_1 : \theta > \theta_0}_{\text{composite}}$$

- Test: reject H_0 in favour of H_1 if:

$$\frac{\hat{\theta} - \theta_0}{\omega} > c$$

- Type I error probability for each $\theta \in \Theta_0$, and its supremum:

$$\begin{aligned} \text{P}(\text{Type I error} \mid \theta) &= \text{P}\left(\frac{\hat{\theta} - \theta_0}{\omega} > c \mid \theta\right), \quad \theta \leq \theta_0 \\ \sup_{\theta \leq \theta_0} \text{P}(\text{Type I error} \mid \theta) &\leq \alpha \end{aligned}$$

- Split $(\hat{\theta} - \theta_0)/\omega$:

$$\frac{\hat{\theta} - \theta_0}{\omega} = \underbrace{\frac{\hat{\theta} - \theta}{\omega}}_{=Z} + \underbrace{\frac{\theta - \theta_0}{\omega}}_{\leq 0 \text{ for } \theta \leq \theta_0}$$

Choice of c , one-sided

- Supremum over Θ_0 , attained at $\theta = \theta_0$:

$$\begin{aligned}
 \alpha &= \sup_{\theta \leq \theta_0} P(\text{Type I error} \mid \theta) \\
 &= \sup_{\theta \leq \theta_0} P(\text{reject } H_0 \mid \theta) \\
 &= \sup_{\theta \leq \theta_0} P\left(\frac{\hat{\theta} - \theta_0}{\omega} > c \mid \theta\right) \\
 &= \sup_{\theta \leq \theta_0} P\left(\underbrace{\frac{\hat{\theta} - \theta}{\omega}}_{= Z \sim N(0,1)} + \frac{\theta - \theta_0}{\omega} > c \mid \theta\right) \\
 &= \sup_{\theta \leq \theta_0} P\left(Z + \underbrace{\frac{\theta - \theta_0}{\omega}}_{\leq 0, \text{ largest at } \theta = \theta_0} > c\right) \\
 &= P(Z > c) \\
 \implies P(Z \leq c) &= 1 - \alpha \\
 \implies c &= z_{1-\alpha}
 \end{aligned}$$

One-sided test and interval

- Size α test of $H_0 : \theta \leq \theta_0$ against $H_1 : \theta > \theta_0$:

Reject H_0 if $\frac{\hat{\theta} - \theta_0}{\omega} > z_{1-\alpha}$

- One-sided interval, all θ_0 the test does not reject:

$$\begin{aligned}
 CI_{1-\alpha}^1 &= \left\{ \theta_0 : \frac{\hat{\theta} - \theta_0}{\omega} \leq z_{1-\alpha} \right\} \\
 &= \{ \theta_0 : \theta_0 \geq \hat{\theta} - z_{1-\alpha} \omega \} \\
 &= [\hat{\theta} - z_{1-\alpha} \omega, +\infty)
 \end{aligned}$$

Mirror case: $H_0 : \theta \geq \theta_0$

- Reject $H_0 : \theta \geq \theta_0$ in favour of $H_1 : \theta < \theta_0$ if, for some $c > 0$:

$$\frac{\hat{\theta} - \theta_0}{\omega} < -c$$

- Choose c from the supremum over $\theta \geq \theta_0$:

$$\begin{aligned}
 \alpha &\geq P(\text{Type I error} \mid \theta) \quad (\theta \geq \theta_0) \\
 &= P\left(\frac{\hat{\theta} - \theta_0}{\omega} < -c \mid \theta\right) \\
 &= P\left(Z + \frac{\theta - \theta_0}{\omega} < -c\right) \\
 \alpha &= \sup_{\theta \geq \theta_0} P\left(Z + \underbrace{\frac{\theta - \theta_0}{\omega}}_{\geq 0} < -c\right) \quad (\text{size } \alpha) \\
 &= P(Z < -c) \quad (\theta = \theta_0) \\
 \implies -c &= z_\alpha \\
 \implies c &= z_{1-\alpha}
 \end{aligned}$$

- Test:

$$\text{Reject } H_0 : \theta \geq \theta_0 \text{ if } \frac{\hat{\theta} - \theta_0}{\omega} < -z_{1-\alpha}$$

Power through Φ

- $\Phi(x) = P(Z \leq x)$, the CDF of $N(0, 1)$.
- Left-tailed test of $H_0 : \theta \geq \theta_0$:

$$\begin{aligned} \pi(\theta) &= P\left(\frac{\hat{\theta} - \theta_0}{\omega} < -z_{1-\alpha} \mid \theta\right) \\ &= P\left(Z + \frac{\theta - \theta_0}{\omega} < -z_{1-\alpha}\right) \\ &= P\left(Z < -z_{1-\alpha} - \frac{\theta - \theta_0}{\omega}\right) \\ &= \Phi\left(\frac{\theta_0 - \theta}{\omega} - z_{1-\alpha}\right) \end{aligned}$$

Two tests for $H_0 : \theta = \theta_0$

- Against $H_1 : \theta \neq \theta_0$, both of size α :

$$\text{Test 1 (one-sided)} : \quad \text{reject } H_0 \text{ if } \frac{\hat{\theta} - \theta_0}{\omega} > z_{1-\alpha}$$

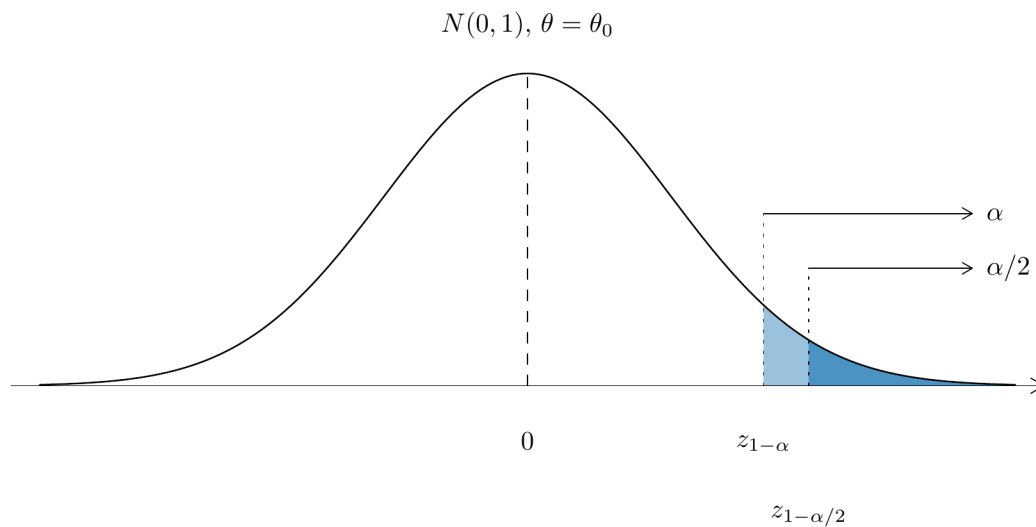
$$\text{Test 2 (two-sided)} : \quad \text{reject } H_0 \text{ if } \left| \frac{\hat{\theta} - \theta_0}{\omega} \right| > z_{1-\alpha/2}$$

- Type I error of Test 1:

$$\begin{aligned} P(\text{Type I error of Test 1}) &= P\left(\frac{\hat{\theta} - \theta_0}{\omega} > z_{1-\alpha} \mid \theta = \theta_0\right) \\ &\stackrel{\theta=\theta_0}{=} P\left(\frac{\hat{\theta} - \theta}{\omega} > z_{1-\alpha}\right) \\ &= P(Z > z_{1-\alpha}) \\ &= \alpha \end{aligned}$$

- Critical values:

$$\begin{aligned} z_{1-\alpha} &< z_{1-\alpha/2} \\ \implies -z_{1-\alpha/2} &< -z_{1-\alpha} \end{aligned}$$



Power of Test 1 and Test 2

- Power of Test 1:

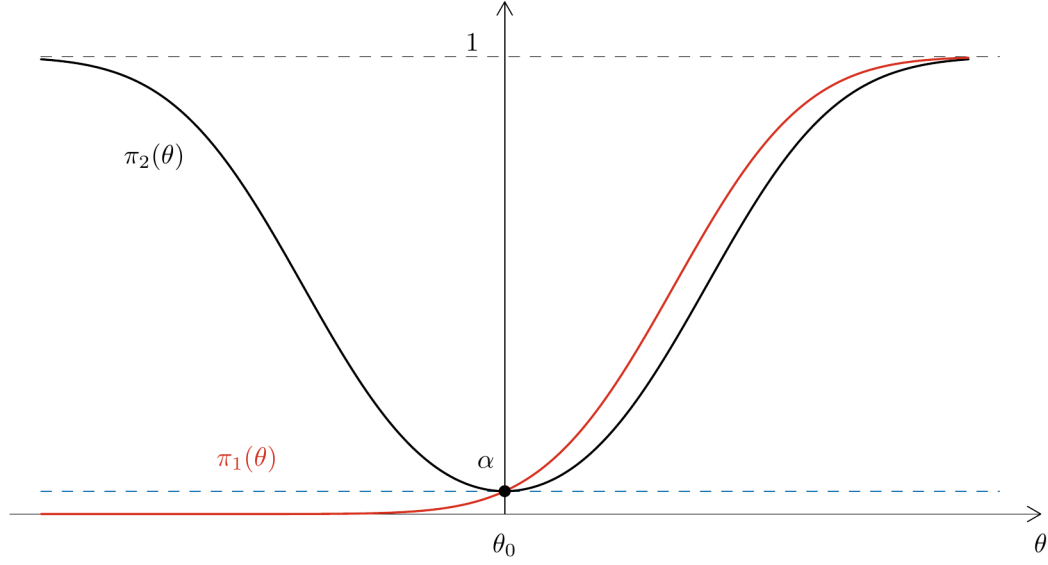
$$\begin{aligned}\pi_1(\theta) &= P\left(\frac{\hat{\theta} - \theta_0}{\omega} > z_{1-\alpha} \mid \theta\right) \\ &= P\left(Z + \frac{\theta - \theta_0}{\omega} > z_{1-\alpha}\right)\end{aligned}$$

- Power of Test 2:

$$\begin{aligned}\pi_2(\theta) &= P\left(\left|\frac{\hat{\theta} - \theta_0}{\omega}\right| > z_{1-\alpha/2} \mid \theta\right) \\ &= P\left(\left|Z + \frac{\theta - \theta_0}{\omega}\right| > z_{1-\alpha/2}\right) \\ &= P\left(Z + \frac{\theta - \theta_0}{\omega} < -z_{1-\alpha/2}\right) + P\left(Z + \frac{\theta - \theta_0}{\omega} > z_{1-\alpha/2}\right)\end{aligned}$$

One-sided against two-sided power

- $\pi_1(\theta)$ increasing, $\pi_2(\theta)$ U-shaped:



- For $\theta > \theta_0$, Test 1 is more powerful; compare the right tails:

$$\begin{aligned} \left| \frac{\hat{\theta} - \theta_0}{\omega} \right| = \frac{\hat{\theta} - \theta_0}{\omega} > z_{1-\alpha/2} \quad \left(\frac{\hat{\theta} - \theta_0}{\omega} > 0 \right) \\ \Rightarrow \frac{\hat{\theta} - \theta_0}{\omega} > z_{1-\alpha} \end{aligned}$$

- For $\theta < \theta_0$, Test 2 is more powerful:

$$\begin{aligned} \frac{\hat{\theta} - \theta_0}{\omega} = Z + \underbrace{\frac{\theta - \theta_0}{\omega}}_{< 0} > z_{1-\alpha} \quad (\text{Test 1 rejects}) \\ \pi_1(\theta) &= P\left(Z + \frac{\theta - \theta_0}{\omega} > z_{1-\alpha}\right) \\ &= P\left(Z > z_{1-\alpha} - \frac{\theta - \theta_0}{\omega}\right) \\ &< P(Z > z_{1-\alpha}) \\ &= \alpha \end{aligned}$$

- No uniformly most powerful test against $H_1 : \theta \neq \theta_0$.
- Test 1 biased against $H_1 : \theta \neq \theta_0$.
- Largest Type II error probability of Test 1:

$$\begin{aligned} \text{against } H_1 : \theta > \theta_0 : \quad \sup_{\theta > \theta_0} (1 - \pi_1(\theta)) &= 1 - \alpha \\ \text{against } H_1 : \theta \neq \theta_0 : \quad \sup_{\theta \neq \theta_0} (1 - \pi_1(\theta)) &= 1 \end{aligned}$$

Two-sided test with unequal tails

- Test 3 of $H_0 : \theta = \theta_0$ against $H_1 : \theta \neq \theta_0$, reject H_0 when:

$$\frac{\hat{\theta} - \theta_0}{\omega} > z_{1-\alpha/3} \quad \text{or} \quad \frac{\hat{\theta} - \theta_0}{\omega} < -z_{1-2\alpha/3}$$

- Type I error probability, with $z_\tau = -z_{1-\tau}$:

$$\begin{aligned}
P(\text{Type I error}) &= P\left(\frac{\hat{\theta} - \theta_0}{\omega} > z_{1-\alpha/3} \text{ or } \frac{\hat{\theta} - \theta_0}{\omega} < -z_{1-2\alpha/3} \mid \theta = \theta_0\right) \\
&= P\left(\frac{\hat{\theta} - \theta_0}{\omega} > z_{1-\alpha/3} \mid \theta = \theta_0\right) \\
&\quad + P\left(\frac{\hat{\theta} - \theta_0}{\omega} < -z_{1-2\alpha/3} \mid \theta = \theta_0\right) \\
&\stackrel{\theta=\theta_0}{=} P\left(\frac{\hat{\theta} - \theta}{\omega} > z_{1-\alpha/3}\right) + P\left(\frac{\hat{\theta} - \theta}{\omega} < -z_{1-2\alpha/3}\right) \\
&= \underbrace{P(Z > z_{1-\alpha/3})}_{=\alpha/3} + \underbrace{P(Z < -z_{1-2\alpha/3})}_{=P(Z < z_{2\alpha/3})=2\alpha/3} \\
&= \alpha
\end{aligned}$$

Power with unequal tails

- Power of Test 3:

$$\begin{aligned}
\pi_3(\theta) &= P(\text{reject } H_0 \mid \theta) \\
&= P\left(\frac{\hat{\theta} - \theta_0}{\omega} > z_{1-\alpha/3} \mid \theta\right) + P\left(\frac{\hat{\theta} - \theta_0}{\omega} < -z_{1-2\alpha/3} \mid \theta\right) \\
&= P\left(Z + \frac{\theta - \theta_0}{\omega} > z_{1-\alpha/3}\right) + P\left(Z + \frac{\theta - \theta_0}{\omega} < -z_{1-2\alpha/3}\right) \\
&= P\left(Z > z_{1-\alpha/3} - \frac{\theta - \theta_0}{\omega}\right) + P\left(Z < -z_{1-2\alpha/3} - \frac{\theta - \theta_0}{\omega}\right)
\end{aligned}$$

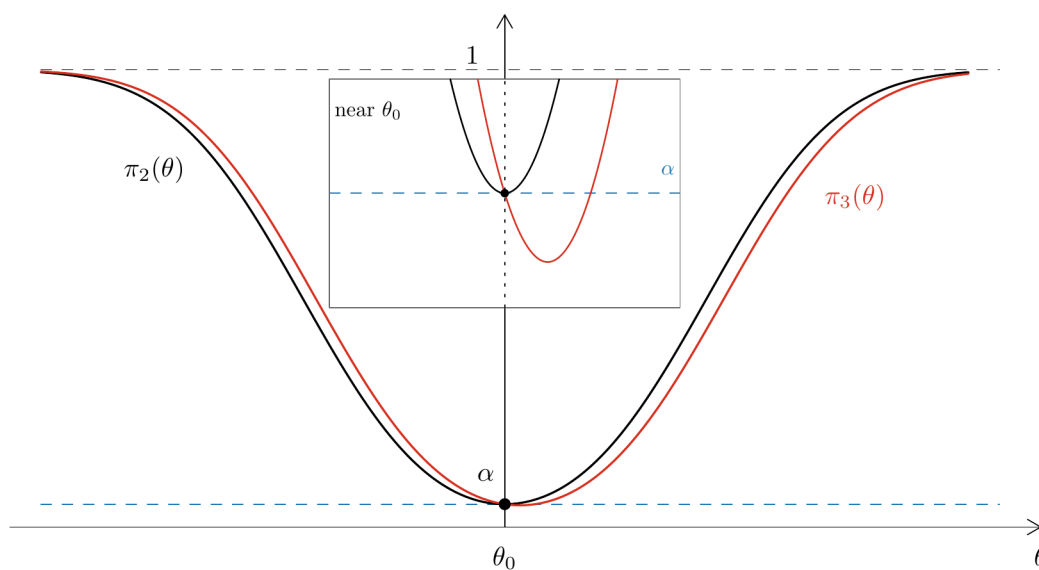
- At $\theta = \theta_0$:

$$\begin{aligned}
\pi_3(\theta_0) &= P(Z > z_{1-\alpha/3}) + P(Z < -z_{1-2\alpha/3}) \\
&= \frac{\alpha}{3} + \frac{2\alpha}{3} \\
&= \alpha
\end{aligned}$$

- For $\theta > \theta_0$, $(\theta - \theta_0)/\omega > 0$:

$$\pi_3(\theta) = \underbrace{P\left(Z > z_{1-\alpha/3} - \frac{\theta - \theta_0}{\omega}\right)}_{>\alpha/3} + \underbrace{P\left(Z < -z_{1-2\alpha/3} - \frac{\theta - \theta_0}{\omega}\right)}_{<2\alpha/3}$$

- $\pi_3(\theta)$ against $\pi_2(\theta)$, both α at θ_0 :



- $\pi_3(\theta) < \alpha$ just right of θ_0 : Test 3 biased against $H_1 : \theta \neq \theta_0$.

Strict inequality in the null

- $H_0 : \theta < \theta_0$ vs $H_1 : \theta \geq \theta_0$, same one-sided test:

$$\begin{aligned} \sup_{\theta < \theta_0} \mathbb{P}\left(Z + \frac{\theta - \theta_0}{\omega} > z_{1-\alpha}\right) &= \sup_{\theta \in (-\infty, \theta_0)} \mathbb{P}\left(Z + \frac{\theta - \theta_0}{\omega} > z_{1-\alpha}\right) \\ &= \mathbb{P}(Z > z_{1-\alpha}) \\ &= \alpha \end{aligned}$$

- $\theta_0 \in \Theta_1$: no power above α at θ_0 .

Null $\theta \neq \theta_0$: not testable

- $H_0 : \theta \neq \theta_0$ vs $H_1 : \theta = \theta_0$; size control:

$$\sup_{\theta \neq \theta_0} \mathbb{P}(\text{reject } H_0 \mid \theta) \leq \alpha$$

- Reject if $|(\hat{\theta} - \theta_0)/\omega| < c$:

$$\begin{aligned} \alpha &\geq \sup_{\theta \neq \theta_0} \mathbb{P}\left(\left|\frac{\hat{\theta} - \theta_0}{\omega}\right| < c \mid \theta\right) \\ &= \sup_{\theta \neq \theta_0} \mathbb{P}\left(\left|Z + \frac{\theta - \theta_0}{\omega}\right| < c\right) \\ &= \mathbb{P}(|Z| < c) \quad (\theta \rightarrow \theta_0) \\ &= \mathbb{P}(\text{reject } H_0 \mid \theta = \theta_0) \end{aligned}$$

- Size α : no power above α at θ_0 , not testable.

p-values

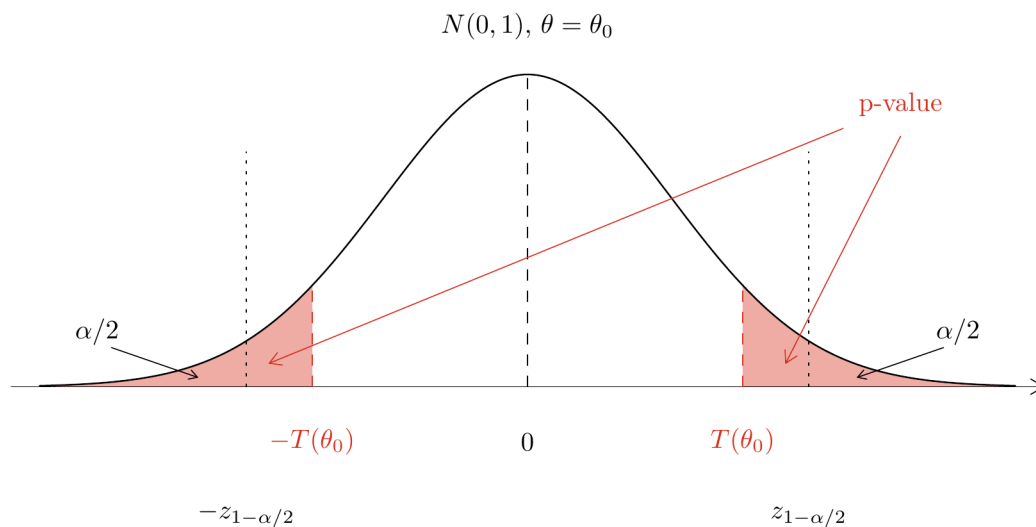
Question

- **Question.** What is the p-value?

1. P(Type I error)
2. Probability of drawing $\hat{\theta}$ from the distribution centred at θ
3. Probability of rejecting
4. Probability of drawing another $\hat{\theta}$ further than the one in your sample

Two-sided p-value

- Two tails beyond $\pm T(\theta_0)$:



- p-value:

$$\begin{aligned}
 \text{p-value} &= 2(1 - \Phi(T(\theta_0))) \\
 &= \boxed{2\left(1 - \Phi\left(\left|\frac{\hat{\theta} - \theta_0}{\omega}\right|\right)\right)} \\
 &= 2\Phi\left(-\left|\frac{\hat{\theta} - \theta_0}{\omega}\right|\right) \quad (1 - \Phi(x) = \Phi(-x))
 \end{aligned}$$

- p-value: random before the data are plugged in, a statistic.
- Test:

$$\text{Reject } H_0 : \theta = \theta_0 \text{ if } \text{p-value} < \alpha$$

- **Claim.** Size α :

$$\begin{aligned}
 \text{P(Type I error)} &= \text{P}(\text{p-value} < \alpha \mid \theta = \theta_0) \\
 &= \text{P}(2(1 - \Phi(|Z|)) < \alpha) \quad \left(\frac{\hat{\theta} - \theta_0}{\omega} = Z \text{ under } \theta = \theta_0\right) \\
 &= \alpha
 \end{aligned}$$

Two-sided p-value under $\theta = \theta_0$

- CDF of the p-value, for $t \in (0, 1)$:

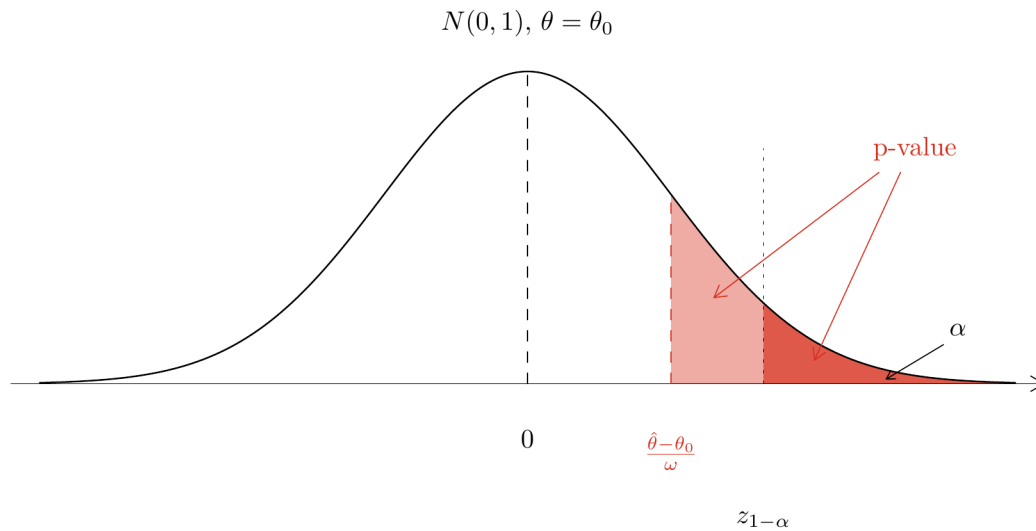
$$\begin{aligned}
 P(\text{p-value} \leq t \mid \theta = \theta_0) &= P\left(2\left(1 - \Phi\left(\left| \frac{\hat{\theta} - \theta_0}{\omega} \right| \right)\right) \leq t \mid \theta = \theta_0\right) \\
 &\quad \quad \quad = \underbrace{Z \sim N(0,1)} \\
 &= P(2(1 - \Phi(|Z|)) \leq t) \\
 &= P\left(1 - \Phi(|Z|) \leq \frac{t}{2}\right) \\
 &= P\left(\Phi(|Z|) \geq 1 - \frac{t}{2}\right) \\
 &= P\left(|Z| \geq \Phi^{-1}\left(1 - \frac{t}{2}\right)\right) \\
 &= P(|Z| \geq z_{1-t/2}) \\
 &= P(Z \geq z_{1-t/2}) + P(Z \leq \underbrace{-z_{1-t/2}}_{=z_{t/2}}) \\
 &= \frac{t}{2} + \frac{t}{2} \\
 &= t
 \end{aligned}$$

- Under $\theta = \theta_0$, for all $t \in (0, 1)$:

$$\begin{aligned}
 &P(\text{p-value} \leq t \mid \theta = \theta_0) = t \\
 \Rightarrow &\boxed{\text{p-value} \sim \text{Uniform}(0, 1)}
 \end{aligned}$$

One-sided p-value, right tail

- Right tail beyond $(\hat{\theta} - \theta_0)/\omega$, for $H_0 : \theta \leq \theta_0$:



- Right-tailed test:

$$\boxed{\text{p-value} = 1 - \Phi\left(\frac{\hat{\theta} - \theta_0}{\omega}\right)}$$

- The same, as a probability over $Z^* \sim N(0, 1)$ independent of the data:

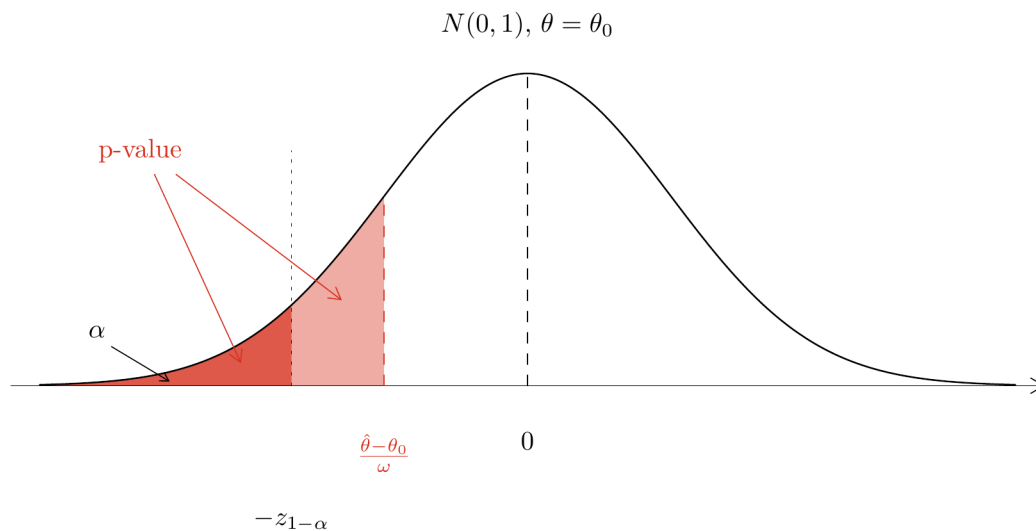
$$\begin{aligned} \text{p-value} &= P^*\left(Z^* > \frac{\hat{\theta} - \theta_0}{\omega} \mid \hat{\theta}\right) \\ &= 1 - \Phi\left(\frac{\hat{\theta} - \theta_0}{\omega}\right) \end{aligned}$$

- Reject H_0 if p-value $< \alpha$.
- Same decision as the one-sided test:

$$\begin{aligned} 1 - \Phi\left(\frac{\hat{\theta} - \theta_0}{\omega}\right) &< \alpha \\ \Leftrightarrow \frac{\hat{\theta} - \theta_0}{\omega} &> z_{1-\alpha} \end{aligned}$$

One-sided p-value, left tail

- Left tail below $(\hat{\theta} - \theta_0)/\omega$, for $H_0 : \theta \geq \theta_0$:



- Left-tailed test: reject if $(\hat{\theta} - \theta_0)/\omega < -z_{1-\alpha}$.
- p-value:

$$\text{p-value} = \Phi\left(\frac{\hat{\theta} - \theta_0}{\omega}\right)$$

- Reject H_0 if p-value $< \alpha$.

Size of the right-tailed p-value test

- We will use: $\Phi^{-1}(1 - \alpha) = z_{1-\alpha}$.

- Supremum over Θ_0 , attained at $\theta = \theta_0$:

$$\begin{aligned}
\sup_{\theta \leq \theta_0} \mathbb{P}(\text{p-value} < \alpha \mid \theta) &= \sup_{\theta \leq \theta_0} \mathbb{P}\left(1 - \Phi\left(\frac{\hat{\theta} - \theta_0}{\omega}\right) < \alpha \mid \theta\right) \\
&= \sup_{\theta \leq \theta_0} \mathbb{P}\left(\Phi\left(\frac{\hat{\theta} - \theta_0}{\omega}\right) > 1 - \alpha \mid \theta\right) \\
&= \sup_{\theta \leq \theta_0} \mathbb{P}\left(\Phi\left(Z + \underbrace{\frac{\theta - \theta_0}{\omega}}_{\leq 0}\right) > 1 - \alpha\right) \\
&= \mathbb{P}(\Phi(Z) > 1 - \alpha) \\
&= \mathbb{P}(Z > \Phi^{-1}(1 - \alpha)) \\
&= \mathbb{P}(Z > z_{1-\alpha}) \\
&= \alpha
\end{aligned}$$

Distribution of $\Phi(Z)$

- New random variable:

$$W = \Phi(Z), \quad W \in (0, 1)$$

- CDF of W :

$$\mathbb{P}(W \leq x) = \begin{cases} 0 & x \leq 0 \\ ? & 0 < x < 1 \\ 1 & x \geq 1 \end{cases}$$

- We will use: $a \leq b \iff g(a) \leq g(b)$ for strictly increasing g .
- For $0 < x < 1$, apply Φ^{-1} inside:

$$\begin{aligned}
\mathbb{P}(W \leq x) &= \mathbb{P}(\Phi(Z) \leq x) \\
&= \mathbb{P}(\Phi^{-1}(\Phi(Z)) \leq \Phi^{-1}(x)) \\
&= \mathbb{P}(Z \leq \underbrace{\Phi^{-1}(x)}_{=z_x}) \\
&= \Phi(\Phi^{-1}(x)) \\
&= x
\end{aligned}$$

- Hence:

$$\mathbb{P}(\Phi(Z) \leq x) = x \implies \boxed{\Phi(Z) \sim \text{Uniform}(0, 1)}$$

Distribution of $1 - \Phi(Z)$

- CDF of $1 - \Phi(Z)$, for $0 < u < 1$:

$$\begin{aligned}
F_{1-W}(u) &= \mathbb{P}(1 - W \leq u) \\
&= \mathbb{P}(1 - \Phi(Z) \leq u) \\
&= \mathbb{P}(\Phi(Z) \geq 1 - u) \\
&= \mathbb{P}(Z \geq \Phi^{-1}(1 - u)) \\
&= 1 - \Phi(\Phi^{-1}(1 - u)) \\
&= 1 - (1 - u) \\
&= u
\end{aligned}$$

- Hence:

$$\mathbb{P}(1 - \Phi(Z) \leq u) = u \implies \boxed{1 - \Phi(Z) \sim \text{Uniform}(0, 1)}$$

p-values under $\theta = \theta_0$

- $(\hat{\theta} - \theta_0)/\omega = Z$:

$$1 - \Phi(Z) \sim \text{Uniform}(0, 1)$$

$$\Phi(Z) \sim \text{Uniform}(0, 1)$$

$$2(1 - \Phi(|Z|)) \sim \text{Uniform}(0, 1)$$

- $(\hat{\theta} - \theta_0)/\omega \sim N(0, 1)$: critical values $z_{1-\alpha}$, $z_{1-\alpha/2}$ from $N(0, 1)$.
- p-value $\sim \text{Uniform}(0, 1)$: critical value α from $\text{Uniform}(0, 1)$.
- For every p-value:

$$\begin{aligned} P(\text{p-value} < \alpha \mid \theta = \theta_0) &= P(\text{p-value} \leq \alpha \mid \theta = \theta_0) \\ &= \text{CDF of the p-value at } \alpha, \text{ under } \theta = \theta_0 \\ &= P(V \leq \alpha), \quad V \sim \text{Uniform}(0, 1) \\ &= \alpha \end{aligned}$$

- In general: $X \sim F$, F continuous and strictly increasing $\implies F(X) \sim \text{Uniform}(0, 1)$.

Summary

Hypotheses, errors, size

- $\hat{\theta} \sim N(\theta, \omega^2)$, ω^2 known; $\theta \in \Theta = \Theta_0 \cup \Theta_1$, $\Theta_0 \cap \Theta_1 = \emptyset$; $H_0 : \theta \in \Theta_0$ vs $H_1 : \theta \in \Theta_1$.
- Test statistic $T \in S = S_A \cup S_R$; reject H_0 if $T \in S_R$.
- Type I error: reject a true H_0 ; Type II error: fail to reject a false H_0 .
- $S_R \downarrow \implies P(\text{Type I error}) \downarrow \implies P(\text{Type II error}) \uparrow$.
- Valid test: $\sup_{\theta \in \Theta_0} P(T \in S_R \mid \theta) \leq \alpha$; size α test: equality.

Two-sided test and power

- Reject $H_0 : \theta = \theta_0$ if $|(\hat{\theta} - \theta_0)/\omega| > z_{1-\alpha/2}$; at $\alpha = 0.05$, $z_{0.975} \approx 1.96$.
- $|\hat{\theta}/\omega| > z_{1-\alpha/2}$: $\hat{\theta}$ significant at level α .
- Fail to reject $\iff \theta_0 \in CI_{1-\alpha}$; $CI_{1-\alpha} = \{\theta_0 : T(\theta_0) \leq z_{1-\alpha/2}\}$.
- $(\hat{\theta} - \theta_0)/\omega = Z + (\theta - \theta_0)/\omega \sim N((\theta - \theta_0)/\omega, 1)$.
- $\pi(\theta) = P(|Z + (\theta - \theta_0)/\omega| > z_{1-\alpha/2})$, $\pi(\theta_0) = \alpha$.
- $\sup_{\theta \neq \theta_0} P(\text{Type II error} \mid \theta) = \sup_{\theta \neq \theta_0} (1 - \pi(\theta)) = 1 - \alpha$.
- Smaller ω : more power.
- Test that ignores the data: size α , power α .

One-sided tests and p-values

- $H_0 : \theta \leq \theta_0$: reject if $(\hat{\theta} - \theta_0)/\omega > z_{1-\alpha}$; $z_{1-\alpha} < z_{1-\alpha/2}$.
- $\sup_{\theta \leq \theta_0} P(\text{reject } H_0 \mid \theta)$ attained at $\theta = \theta_0$.
- $CI_{1-\alpha}^1 = [\hat{\theta} - z_{1-\alpha}\omega, +\infty)$.
- Against $H_1 : \theta \neq \theta_0$: one-sided test biased; no uniformly most powerful test.

- Unequal tails, $\alpha/3$ right and $2\alpha/3$ left: size α , biased against $H_1 : \theta \neq \theta_0$.
- p-values; reject if p-value $< \alpha$:

two-sided: $2(1 - \Phi(|(\hat{\theta} - \theta_0)/\omega|))$

right-tailed: $1 - \Phi((\hat{\theta} - \theta_0)/\omega)$

left-tailed: $\Phi((\hat{\theta} - \theta_0)/\omega)$

- Under $\theta = \theta_0$: every p-value $\sim \text{Uniform}(0, 1)$; $P(\text{p-value} < \alpha \mid \theta = \theta_0) = \alpha$.